

Literature Review

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1 Introduction

In astronomy, the group of objects that make up temporary phenomena such as supernovae (SNe), gamma ray bursts and tidal disruption events are known as transients. The observation of transients is critical in astronomy and astrophysics, as the data can lead to important discoveries such as the accelerating expansion of the Universe (Perlmutter et al., 1999; Riess et al., 1998). SNe are particularly important due to a specific type of supernova (SN) called type Ia supernovae (SNe Ia).

When SNe Ia are observed, their brightness can be plotted as a function of time creating what is known as a light curve. Using the Tripp relation with empirical fitting models like SALT the light curves can be standardised, which is to correct for numerous factors that affect where the peak magnitude occurs in the evolution of the light curve, to a common peak magnitude with some remaining scatter (Ginolin et al., 2025; Kelsey et al., 2022; Martin et al., 2024; Tripp, 1998). Standardisation enables the ability to find the distance to the SN. Using these distance calculations, the parameters that describe the properties of the Universe can be determined.

However, the current methods of standardisation may not be comprehensive evidenced by remaining residuals correlating with host galaxy dust and mass. If these effects can be accounted for, then truly representative light curves of these SNe can be used to produce more precise cosmological calculations.

2 Supernova Ia physics and Progenitor Channels

2.1 Standardisation

SNe Ia are standardisable candles, which means that SN Ia are SN Ia are powered similarly, so their peak brightness is similar. By accounting for correlations with SN Ia light curve shape and colour, their brightness can be corrected to reach <10% dispersion and thus standardisation. The corrections for the light curves can be found in equation 1 which accounts for the colour and stretch of the supernovae to determine the distance modulus (μ_{obs}) (Tripp, 1998). Using this distance modulus you can then derive distances to the

Supernovae.

$$\mu_{obs} = m_B^* - M_B + \alpha x_1 - \beta c \tag{1}$$

In equation 1 the distance modulus is the difference between the apparent magnitude, m_B^*, of the supernova and the absolute magnitude, M_B of the supernova. m_B^* specifically represents the peak apparent magnitude of the SN in the rest frame blue filter of the observing instrument. The blue band is an imaginary band that occupies the wavelength range of blue optical light and due to the effects of cosmological redshift, the blue light from SN are typically observed in the green wavelength range. The remaining parameters are the stretch x_1 and the colour c where α and β are the stretch and colour coefficients, known as nuisance parameters that vary depending on the properties of the sample (Brout et al., 2022; Martin et al., 2024). In modern studies of classification of SNe Ia, an empirical model spectral energy distribution model can be used to obtain m_B^*, x_1 and c, the original SALT model developed by Guy et al. (2005) was trained on 50 light curves of known SNe Ia and can be used to derive the standardisation parameters, x_1 and c, described in equation 1. Further improvements have been made to the SALT model for light curve fitting with SALT3 being the most recent and having improved performance in stretch and colour relations and improves the estimation of uncertainties(Kenworthy et al., 2021).

x_1 corrects for the width of the light curve due to different distributions and masses of ^{56}Ni in the ejecta of the supernova (Deckers et al., 2022). Whilst c is describing the colour of the SN that is observed and contributes to standardisation by describing the environment of the SN. In dusty environments the shorter (bluer) wavelength light gets scattered more which results in a SN that appears more red in colour, this alters the shape of the light curve (Kelsey et al., 2022). In general, the relationship described by Tripp (1998) is that brighter SNe last longer and are bluer than dimmer SNe. Finally, m_B^* is the apparent magnitude of the SNe Ia which is how bright the object appears from Earth and M_B is the absolute magnitude of the SNe Ia which is how bright the SNe Ia is from 10 parsecs away.

The standardisation of SNe Ia light curves is not perfect, with the Tripp equation still leaving enough scatter of the distance modulus that is not accounted for through the un-

certainty of data collection process (Brout et al., 2022; Martin et al., 2024; Rigault et al., 2020; Wiseman et al., 2023). In Figure 1 a residual correlation can be observed showing how the average for the Hubble residual, decreases in SNe that happen in galaxies that have a log stellar mass greater than 10(Sullivan et al., 2010). The Hubble residual is the measure of scatter in the calculation of the distance modulus, it's determined by the subtraction of a distance modulus calculated from a specific cosmology, μ_{cos}, from μ_{obs}, seen in equation 2. Research continues to determine how to improve standardisation methods that result in reducing correlations which will therefore determine cosmological parameters that better represent the Universe. It has been found that the scatter after corrections using the Tripp equation is reduced to 0.15 mag, leading to the conclusion that further corrections based on different properties are needed to standardise these light curves (Brout et al., 2022; Kelsey et al., 2022; Martin et al., 2024).

$$\Delta\mu = \mu_{obs} - \mu_{cos} \tag{2}$$

A contentious correction that is made after the corrections made in equation 1 to the SNe Ia light curves, is that of the earlier mentioned host galaxy stellar mass step(Kelly et al., 2010; Lampeitl et al., 2010; Sullivan et al., 2010). This remaining correlation showed that SNe Ia are 10 per cent brighter in more massive host galaxies.

Kelly et al. (2010), Lampeitl et al. (2010), and Sullivan et al. (2010) independently found the correlation of Hubble residuals with the stellar mass of the SN's host galaxy and provided the initial mass step correction to further standardise supernova light curves. The aforementioned studies demonstrate how a correlation like the mass-step is found across a variety of redshifts up to 0.85 across multiple surveys and has become an important correlation that has been attempted to be corrected for. The physical mechanics of why this difference in brightness is observed across more massive galaxies is still unknown.

2.2 Progenitor Channels

It is accepted that the star that undergoes a supernova, producing the characteristic SNe Ia light curve is a carbon-oxygen (C-O) white dwarf (Hoyle & Fowler, 1960). The two popular

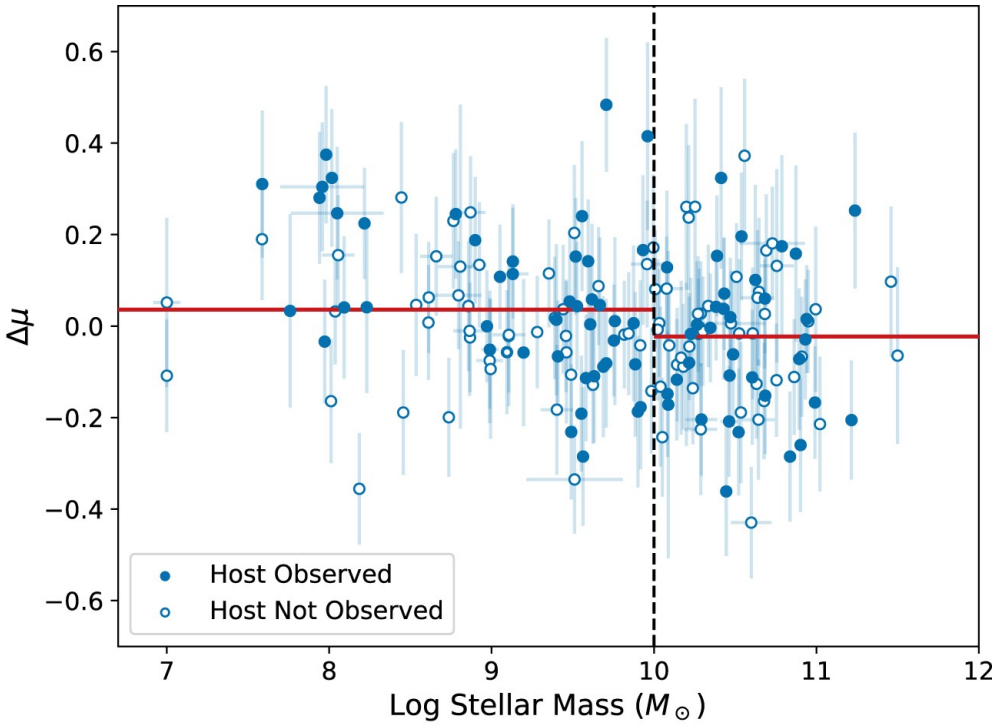


Figure 1: A figure taken from Martin et al. (2024) that shows the scatter in Hubble residuals of 179 SNe Ia from the Foundation Survey dataset as a function of the host galaxy's stellar mass. The average Hubble residual is modelled as a step function, with the 'step' occurring at a host galaxy stellar mass of $10M_\odot$.

progenitor channels for these SNe Ia are the single degenerate scenario (SD) and the double degenerate scenario (DD). SD proposes that before an SN Ia event occurs, the system that was there before consisted of a C-O white dwarf that was accreting material from a non-degenerate companion star in a binary orbit (Whelan & Iben, 1973). Whilst the DD scenario presents the idea of a merger between two degenerate stars such as two C-O white dwarfs and can also result in a supernova(Iben & Tutukov, 1984).

The SD and DD progenitor models have been popular for decades and with recent research indicating that there is more nuance to these two progenitor channels that can cause the variety of the light curves observed by astronomers. Importantly, the SD scenario is commonly associated with the C-O white dwarf reaching the Chandrasekhar mass limit ($M_{Ch} = 1.44M_{\odot}$) whereas the DD scenario could potentially explain the occurrence of Sub M_{Ch} and super M_{Ch} explosion models. To constrain the occurrence of these two scenarios, physical models such as those used in the study done by Deckers et al. (2022).

Deckers et al. (2022) shows that out of 115 SNe Ia events that were detected by the Zwicky Transient Facility (ZTF), 67 per cent of the observed supernova light curves could effectively be recreated by the TURTLS model where the brightness of the explosions are a direct result of Chandrasekhar-mass explosions (progenitors with $M = 1.44M_{\odot}$) with extended distributions and masses of 0.4 to 0.6 $M_{\odot}$ of ^{56}Ni. Deckers et al. (2022) shows that over two thirds of SNe Ia could be resulting from Chandrasekhar mass C-O white dwarf explosions, whilst also demonstrating that 11 per cent of the SNe Ia light curves in the sample aren't reproduced from Chandrasekhar mass explosions and 19 per cent of the sample cannot be constrained using the varied masses and distributions of ^{56}Ni.

Another study done by Burke et al. (2022) shows that using a model with a companion shock-heating component, where the mass ejected by the supernova collides with the companion and heats up that surface, on a sample of nine SNe Ia with good early light curve data, are well fit by the companion interaction with the ejecta. This modelling adds weight to the SD scenario being the more common progenitor.

Although light curve fitting and modelling has shown results in determining the possible progenitors systems of SNe Ia, there have been numerous studies on the correlations of SNe Ia's and their host galaxy's properties. Kelsey et al. (2022) finds through the analysis of 675

SNe Ia from the Dark Energy Survey 5-year sample, that blue ($c < 0$) SNe Ia are found in host galaxies that have lower stellar mass with higher star formation rates (SFR) and are more blue in colour. Another study, although older is Anderson et al. (2015) which found a similar trend with SNe Ia light following the proportion of the light produced by the more blue regions of the host galaxy. Anderson et al. (2015) asserts that because the light from SNe Ia traces out the bluer region of host galaxies with lower SFR, that progenitor systems that are less than 100 Myr old are extremely rare or do not exist and instead the progenitor system of SNe Ia drifts outward from the centre of the host and then explodes. Coming back to Kelsey et al. (2022), who also find this 'colour gradient' where star formation begins closer to the centre of the host galaxy and eventually spreading to the outskirts of the host. To conclude, Anderson et al. (2015) found that the more 'local' lower star formation rates host bluer SNe Ia, leading to older progenitor systems producing SNe Ia. And Kelsey et al. (2022) finds that the colour of the SNe Ia as a function of the local host galaxy environment (which Kelsey et al. (2022) defines as 4 kpc radially from the SN Ia) colour is more blue in locally blue host environments, both these findings would suggest that the progenitor systems of SNe Ia are old and happening in bluer environments suggesting that the progenitor drifts outwards from the centre of the host.

2.3 Early Excess in SNe Ia

Early excess (EEx) in the light curves of SNe Ia are seen as bumps in the light curve before the light curve of the SNe Ia has reached its peak. In 2012 a SN Ia was observed and the analysis of the light curve by Marion et al. (2016) showed an EEx or bump in the V, B and U bands of the SN. This was the first analysis of a 'normal' SNe Ia that had an EEx and the conclusion drawn was that the progenitor system likely was a non-degenerate companion, that was shock heated and was fortunate enough to have the right viewing angle that the heated surface of the companion was emitting in the line of sight of the observers.

Coming back to Burke et al. (2022), who using data which was obtained from high cadence follow up observations found 9 SNe Ia that had good early light curve data. Burke et al. (2022) found that the light curves were best explained via the shock heating of a non-degenerate companion where a C-O white dwarf was accreting mass from this companion and

upon explosion, had the ejecta collide with the companion star that results in the companion, at the right angle, emitting a near-U, V and blue excess which is predicted by Kasen (2010). This again corroborates with the discovery Marion et al. (2016) proposes, that companion interactions with a non-degenerate star is the physical cause of the EEx observed and is rare due to the viewing angle of the companion not aligning with observations often.

Compare the two previous studies to Deckers et al. (2022), who demonstrates a method to determine EEx in 6 light curves of the SNe Ia in the sample, Deckers et al. (2022) finds that these flux excesses are generally well produced by the non degenerate companion interaction models as well as the ^{56}Ni clumping models. Due to the limited sample size Deckers et al. (2022) concludes that SNe Ia likely have more nuance to explosion than what is known, hence the varied observed light curves. Deckers et al. (2022) concludes that with surveys of the sky that revisit chosen sites more rapidly to collect earlier data on the explosion that the nuance could be derived from that earlier data.

Deckers et al. (2022) argues that the EEx in SNe Ia is heterogeneous and can be caused by ejecta interacting not just with the companion but with other aspects of the environment too. Deckers et al. (2022) and Marion et al. (2016) and Burke et al. (2022) also all add that to be more definitive regarding the mechanisms behind the EEx, one day cadence surveys followed by fast spectroscopic follow up will grant the ability to make more observations of SNe Ia which will result in more EEx SNe Ia being found enabling more conclusive study of the mechanisms of EEx.

Lim et al. (2023) is a case study on SN 2021hpr which exhibits an EEx, similar to Marion et al. (2016) and Burke et al. (2022), the shock heating emission of the companion star for the white dwarf most suitably models the light curves of SN 2021hpr although this time in the B, V, R and I bands from the Intensive Monitoring of Nearby Galaxies Survey which can be seen in Figure 2. Although the light curve fit well with a non degenerate companion model, Lim et al. (2023) could not find evidence of any stripped mass of the companion that would've been expected following such an interaction, a detail that was not present in Burke et al. (2022) and Marion et al. (2016) who made the same conclusion.

Lim et al. (2023) shows that the light curve deviates from a simple power law model as well as the companion model from Kasen (2010) and has a bump across all bands which can

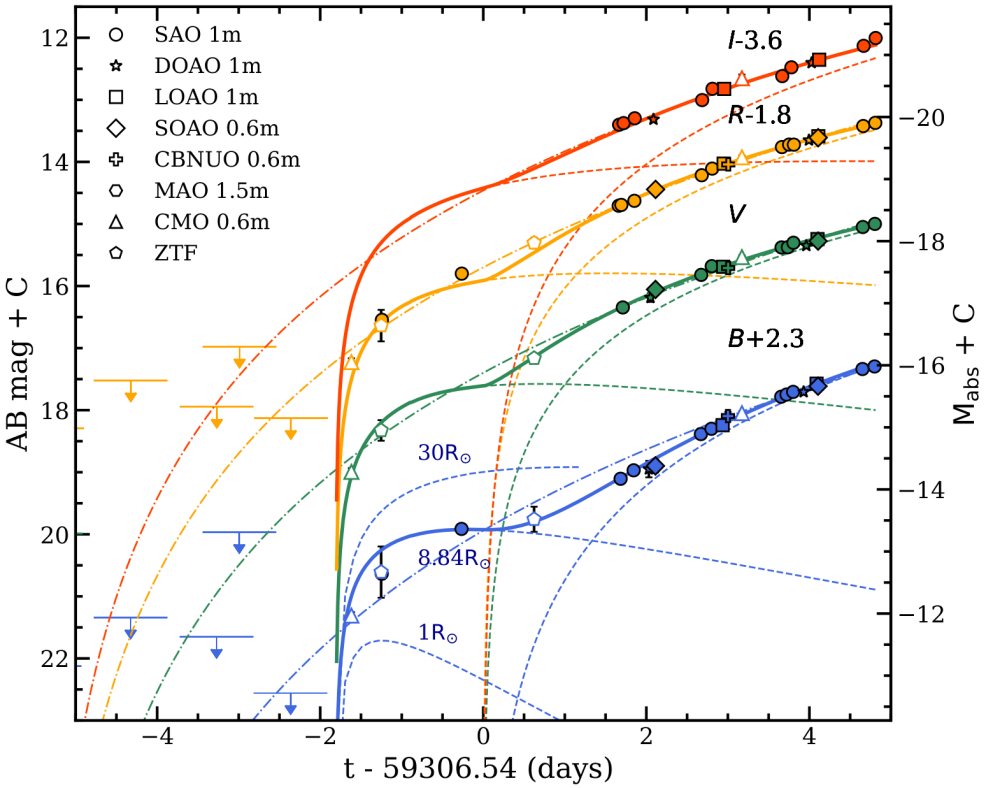


Figure 2: The light curves of SN2021hpr in magnitude space taken from Lim et al. (2023). The points are the observations, the dashed-dotted line is the light curve modelled by the simple power law model and the dashed lines are modelled light curves from Kasen (2010) which include the emission of light from the shock heating of a companion star. Finally, the solid lines are the two models combined and can be seen to line up well with the observations.

be seen in Figure 2. But the combination of both the power law and the companion models from Kasen (2010) produce a curve that fits the observations.

The method used to detect the EEx fluxes in the SNe Ia data can vary, in Deckers et al. (2022) they use the residuals in the light curves that are found by subtracting the modelled light curve from an observed light curve. The baseline modelled light curves are produced by the TURTLS model(Magee et al., 2018) which is a 1 dimensional Monte-Carlo simulation that models the light curves of SN that are driven by radiative transfer and incorporates distributions of ^{56}Ni in the modelling. If a positive residual is found in the light curve within the calculated 6 days after the explosion epoch Deckers et al. (2022) classify that SNe Ia light curve as an EEx SNe Ia.

In Burke et al. (2022) the EEx flux SNe Ia are identified in a similar way to what was done in Lim et al. (2023), as Burke et al. (2022) uses the same companion shock cooling models present in Lim et al. (2023) from Kasen (2010). Burke et al. (2022) stipulates that they classify SNe Ia as EEx SNe Ia if there is data available within five days of the first detection, the best fit viewing angle is less than 90 degrees and the residuals in the g band show a systematic decreasing from the earliest epochs to then matching the modelled light curve. Finally in Marion et al. (2016) the EEx flux SNe Ia was detected through the comparison of the observed light curve to the fitted power model light curve where the observed data had to be three times the measurement uncertainty.

The study of EEx SNe Ia is relatively new and with such limited data available to study these object en mass, it is difficult to understand the cause, although current studies show that interactions between the ejecta and the companion is the leading interpretation. But without studies using larger populations, some doubt remains on the cause.

3 Host Galaxy Environment and SN Ia diversity

3.1 Host Galaxy Properties

In section 2.2 it's clear that the environment of the SNe plays a factor in the shape of the light curve of a SN. Thus the properties of the host galaxies of SNe have been studied to determine

the correlations between those properties and where SNe happen. Toy et al. (2025) uses a sample of 1533 SNe Ia from the 'five-year' sample from the DES SN program. A metric called the the directional light radius (DLR) of a host galaxy is used to associate SN with a host galaxy. The DLR represents the radius of the host galaxy's ellipse in the direction of the SN. Toy et al. (2025) uses the DLR to associate SNe Ia with a host galaxy and then finds the normalised DLR which is the ratio of the separation from the host galaxy's centre to the SN and the DLR. Toy et al. (2025) finds that there is no step-like change in the Hubble residuals average, across a normalised DLR, d_{DLR}, of 1, this can be seen in Figure 3. But when splitting the sample to 'inner regions' where $d_{DLR} < 1$ the Hubble residuals have a step in the averages of 0.100 ± 0.014 mag when split across the host galaxies with a stellar mass of 10. But in the outer regions ($d_{DLR} > 1$) Toy et al. (2025) finds that the step in the Hubble residuals with the same host mass split is 0.036 ± 0.018. From this analysis, Toy et al. (2025) concludes that SNe Ia in the outer regions are less affected by the global host galaxy properties, although Toy et al. (2025) notes that the different mechanics that produce this result is still unknown.

By searching for these correlations with the host galaxy properties, an intrinsic property of the SNe Ia's environment could explain this observation of SNe Ia in higher mass galaxies being brighter than those in lower mass galaxies. As seen in Toy et al. (2025), the outer regions of the host galaxy appear to be less affected by whatever is changing the observed brightness of these SNe Ia.

Another interesting property to study is the global colour of the Host galaxy, Kelsey et al. (2022) finds that blue SNe Ia (SNe with $c < 0$ are defined as blue and $c > 0$ as red in colour) have higher homogeneity in the Hubble residuals. This higher homogeneity in blue SNe Ia was found across the global host galaxy mass, local host galaxy mass, global host galaxy colour and local host galaxy colour. Kelsey et al. (2022) interprets this stronger homogeneity, post-standardisation, as a sign that the older, more globally red galaxies have had more time for SNe to occur and create dust that reddens the SNe Ia that happen afterwards, meaning that the red population is a heterogeneous one and could be made up of a different sub type of SNe Ia that needs to be standardised in a different way to reduce the step function in the residuals.

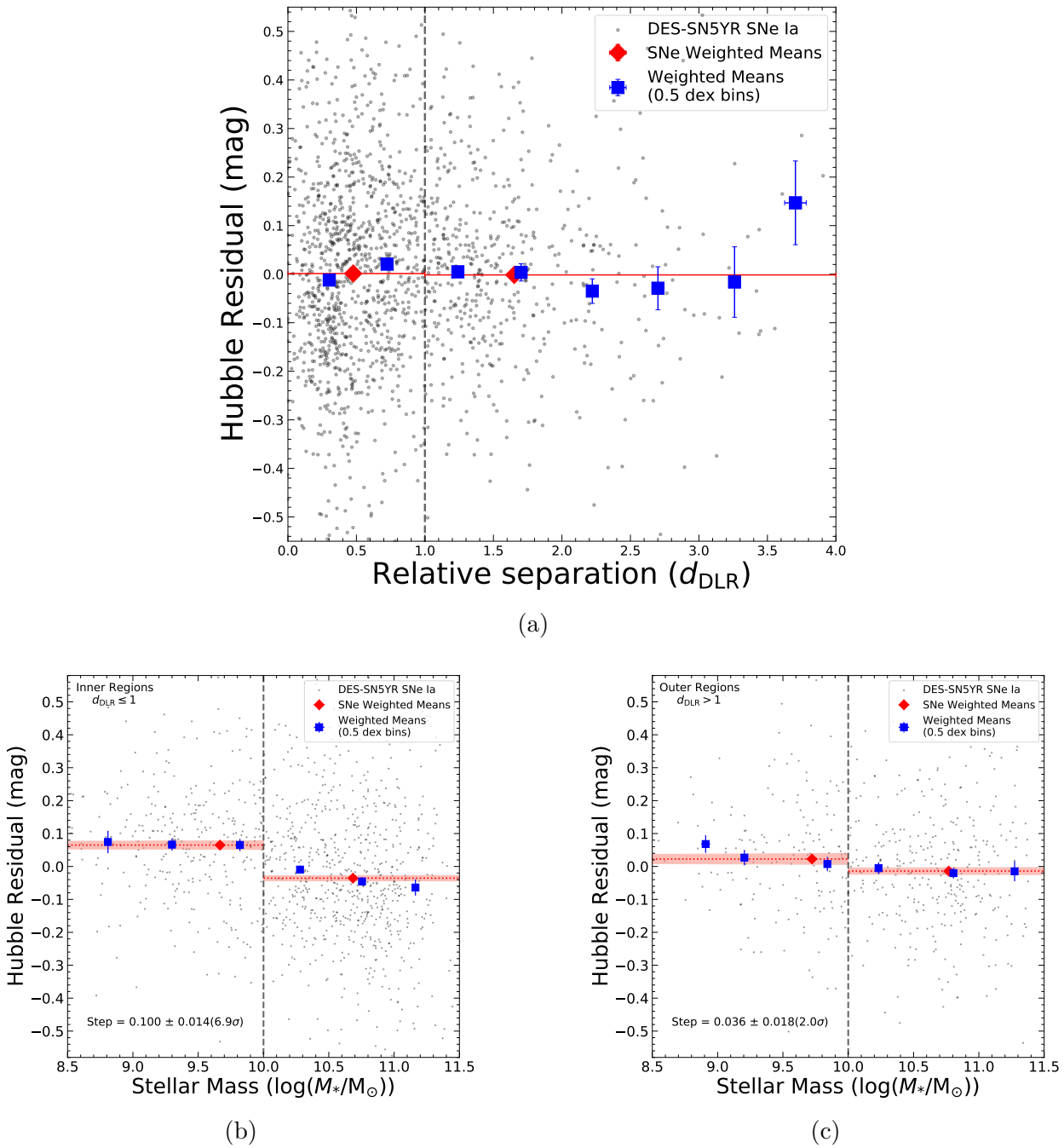


Figure 3: Taken from Toy et al. (2025), Figure (a) shows the Hubble residuals as a function of the relative separation from the galaxy centre to the location of the SNe Ia. As a function of the relative separation, it's obvious that the step between Hubble residuals is greatly reduced. In images (b) and (c) which show the Hubble residuals for the inner regions and outer regions respectively, the mass step is found to be reduced for the SNe Ia in the outer regions of their host galaxies.

Another study, Ramaiya et al. (2025), finds correlations with SNe Ia and the stage of evolution that the host galaxy of the SNe Ia. Ramaiya et al. (2025) uses 1685 SNe Ia from the DES SN sample and for the host galaxies Ramaiya et al. (2025) uses a wide range of sources which include the Canada-France-Hawaii Telescope, the VLT Survey Telescope, the Subaru Hyper Suprime-Cam, the Visible and Infrared Survey Telescope for Astronomy and the Spitzer Space Telescope and the Herschel Space Observatory. This number of sources lets Ramaiya et al. (2025) cover a wide range of light emitted from host galaxies for a robust study. As can be seen in Figure 4, the colour nuisance parameter β changes based on the stage of evolution that the host galaxy is in. Ramaiya et al. (2025) shows that the stage of the host galaxy's evolution which changes properties such as mass and SFR also affects the brightness of the SNe Ia in their host galaxies.

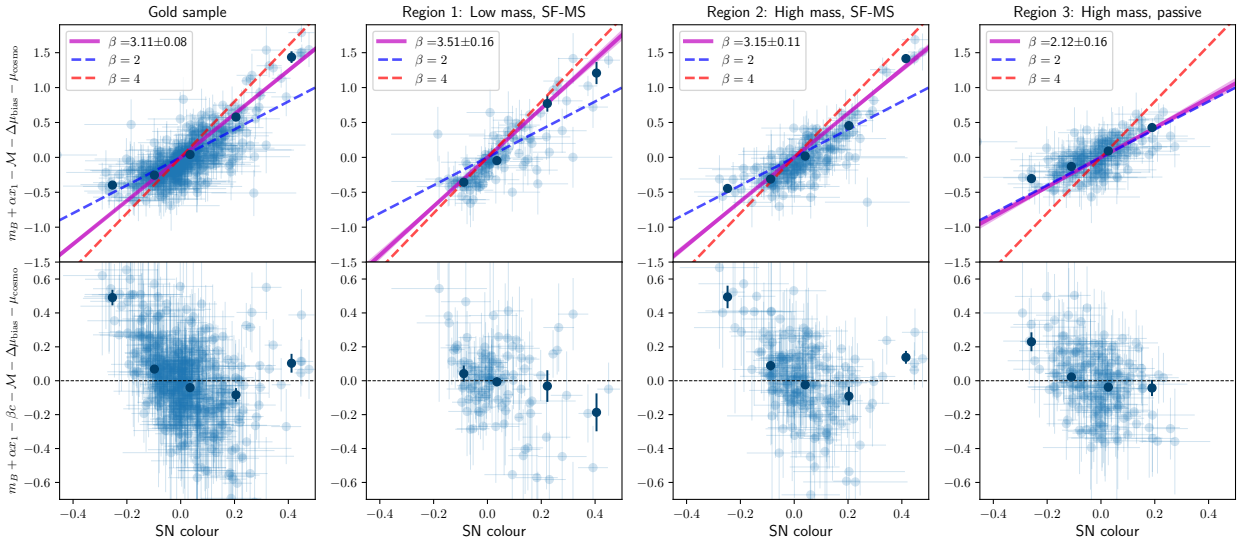


Figure 4: Taken from Ramaiya et al. (2025), the plots on top are the Hubble residuals plotted as a function of SNe Ia colour, without the colour calibration term in the Tripp equation. The bottom plots show the Hubble residuals whilst correcting for colour as a function of the SNe Ia colour. The gold sample is the sample of all the SNe Ia from the DES SN sample, whilst the remaining three are the plots of the gold sample divided by the evolutionary stage the host galaxy is in.

Connecting both Ramaiya et al. (2025) and Kelsey et al. (2022) shows that the evolutionary stage of the host galaxy and ergo the age has an effect on the magnitudes of the SNe Ia. In Ramaiya et al. (2025) β is noticeably different to the active star forming galaxies but once star formation ceases, the colour correction isn't as aggressive. Kelsey et al. (2022)

finds that hosts with a higher $U - R$, which is a proxy for a younger star forming galaxy, host SNe Ia that require a higher β which is also found in Ramaiya et al. (2025). The same relation for more blue galaxies, where $U - R$ is lower requires smaller β. This corroboration of an age-mass relation of host galaxies in Ramaiya et al. (2025) and Kelsey et al. (2022) shows that the age and evolutionary stage of the host could relate to the SNe Ia in a way that affects the brightness of the SN in the galaxies.

3.2 The Environment and Early Excess Flux

Host galaxy correlations are clearly important in the study of SNe Ia, as understanding the environment where SNe Ia are occurring in can help explain the physics of the SN. The correlations with host galaxy mass, DLR, colour and SFR have helped navigate what could be causing the remaining scatter in the Hubble residuals. But what correlations with host galaxy properties are there for EEx SNe Ia and could they help explain the additional bump in the light curve?

Due to how small the sample sizes are for EEx SNe Ia, the studies of host galaxy properties correlating with properties of the EEx SNe Ia are limited. Deckers et al. (2022) found, whilst acknowledging the small sample size doesn't allow them to make significant conclusions, that 5 of the 6 EEx SNe Ia have hosts with stellar masses lower then the mean of their sample $< \log(\mathrm{M}_*/\mathrm{M}_\odot) = 9.82$. Because of the connection to lower mass hosts, Deckers et al. (2022) postulates, stemming from Rigault et al. (2020) that the progenitor systems of EEx SNe Ia are likely to be from a younger population.

A recent study by Wu et al. (2026) uses a sample of 16 SNe Ia with data 16 days before the B band maximum light. Wu et al. (2026) found 3 EEx Flux SNe Ia and could only associate 2 with a host galaxy, both the associated hosts of the 2 EEx SNe Ia had a lower stellar mass than the mean of the sample ($\log(\mathrm{M}_*/\mathrm{M}_\odot) = 9.94$). Whereas one of the SNe Ia occurred in a host with the highest SFR of the sample ($3.64\mathrm{M}_\odot yr^{-1}$) and the second SNe Ia occurred in a galaxy with the fourth lowest SFR of the sample.

This leads to the primary gap in the literature, where the studies of EEx SNe Ia have been with such a limited population that their relationship with their host galaxies properties has remained a mystery. Previous literature has shown that SNe Ia population studies with

host galaxy properties have been able to constrain the progenitors somewhat and can lead to a greater understanding of the explosion physics and supernova environment which can lead to better light curve standardisation.

4 Fink, Rubin and LSST

Over the course of this review it has become obvious that more early data would provide the next steps for both progenitor channel and correlation constraining of SNe Ia with host properties. If more SNe Ia in hosts can be collected with earlier data before peak brightness, then the correlations with host galaxy properties can be studied with a more complete picture of the supernova with the earlier light curve data.

The EEx SNe Ia are difficult to find, with most studies not being population studies but rather case studies, and the few population studies result in a sample size of 6 and 3 (Burke et al., 2022; Deckers et al., 2022; Wu et al., 2026). The Vera C. Rubin Observatory which will be referred to as Rubin from here on out, is located on the El Peñón peak of Cerro Pachón in Chile (Bianco et al., 2022). The various band passes of the camera on the telescope used in the Rubin Observatory range from 300 nanometres to 1100 nanometres, the range of each filter can be seen in Figure 5.

The cadence of LSST will be in general, 2 - 4 days across all the sites in the southern sky. But to be specific, for the Wide-Fast-Deep (WFD) survey an instance of imaging will be followed up by another instance roughly 33 minutes later but with a different filter, which will be followed by a third observation in a different filter 3 to 7 hours later("Vera C. Rubin Observatory Survey Strategy", n.d.). Other portions of the survey such as the Deep Drilling Fields (DDF) will have a much higher cadence. Although without an specified cadence at this stage, the sites will receive 20000 revisits over the 10 year lifetime of LSST which is much higher in comparison to the other survey regimes. These sites and the number of times they will be visited can be seen in Figure 6.

An alert is the notification that a transient candidate has been detected and once LSST is fully underway, 10 million alerts are expected to be produced each night of viewing (Möller et al., 2021). Of these alerts, it is expected that there will be $\approx$ 1100 SNe per night. This

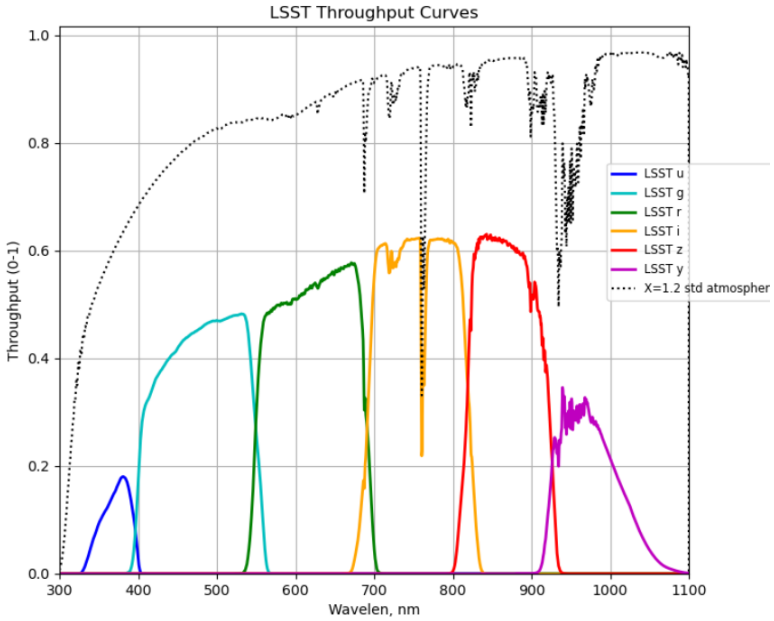


Figure 5: The band passes of the LSST camera, the throughput is the percentage of expected transmission of the observations that will be made by the respective band. Each colour follows the nomenclature for the photometric system where u is near-UV, g is across the blue and green range, r is the red and includes the visual spectrum range, finally i, z and y all make up different bands in the near-infrared ("The LSST Camera (LSSTCam)", n.d.).

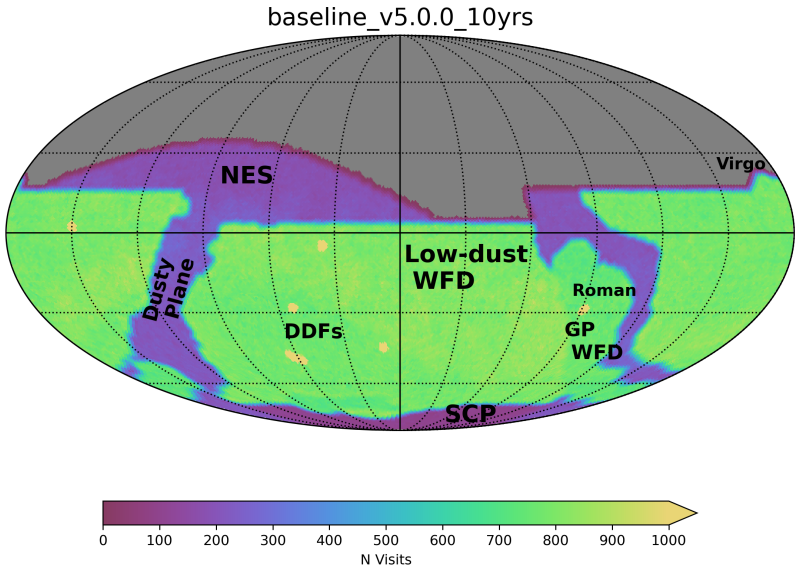


Figure 6: The baseline number of visits that Rubin will make across the Southern Sky. Taken from "Vera C. Rubin Observatory Survey Strategy" (n.d.), the different areas on the map show the main survey, noting the difference between the number of revisits of the yellow DDF and the green WFD. It is easy to see that different surveys using the Rubin Observatory require different viewing cadences to enable the collection of data that is desired.

should create the opportunity to discover more SNe Ia which will increase the odds of finding the EEx SNe Ia and could enable the beginnings of a population study with respect to their host galaxy properties.

To handle this number of alerts, The Fink broker will be instrumental as it enables the processing of the large number of alerts from the alert stream of Rubin. Fink performs the ingestion of the data of the LSST alert stream, annotation of the alerts such as early light curve classification and is capable of customisable filtering of the alerts to allow users to cater the alerts that are recieved to their needs (Möller et al., 2021).

Fink employs machine learning methods of classification to disentangle the types of SNe that will be discovered by Rubin. These machine learning classifiers use an active learning approach that can take a smaller sample of training data and continually retrain the classification models to improve the models as the survey progresses in time as well as other methods(Leoni et al., 2022). This means that SNe Ia candidates will be identified from the LSST alert stream producing science-ready images in mere minutes.

The combination of Rubin, LSST and Fink are critical in for the next generation of

studies on SNe Ia, the volume of data collection is unprecedented with Fink allowing for scientists to easily work with data readily minutes after the alerts from Rubin are sent out (Möller et al., 2021).

5 Project Aims

Here the aims of the project are described, firstly to construct a sample of extra-galactic transients with the Fink broker, using data collected from Rubin. From the sample of extra-galactic candidates, SNe Ia candidates will be identified using the machine learning classifiers in Fink and rise and fall times (Bazin et al., 2011; Leoni et al., 2022; Möller & de Boissière, 2020). The supernova candidates will then have their respective properties derived such as colour and stretch parameters using SALT3. Using the fitted light curves modelled by SALT3, the selection of EEx candidates will occur by finding residuals in the magnitudes between the observed light curve and modelled light curve. Finally, host galaxy properties of the EEx SNe Ia will then be derived using existing methods and correlated with the properties of the EEx SNe Ia.

6 Planned Methodology

The proposed method for this project will be to use alerts from Rubin to find potential extragalactic SN candidates. One of the first quality cuts that will be made is using only 5 sigma alerts, where the flux received by a signal is 5 times greater than the uncertainty of the flux, produced by Rubin which will then be ingested by the Fink broker (Möller et al., 2021). Using the Fink broker, these alerts will be passed through an extragalactic candidates filter, the filter will pass those alerts which have a higher signal to noise ratio, are not already known transients, which is done through the cross matching services in Fink with the Transient Name Server and SIMBAD for example. The extragalactic candidates filter will cut alerts that are within the galactic plane, so that alerts with a galactic latitude of $|b| > 20$ pass the filter giving a sample of extra-galactic transient candidates (Möller et al., 2021). Finally transients that have extended observed histories need to be cut, so any

transient with data that is older than 150 days is cut from the sample (Möller et al., 2021).

Once a sample of extragalactic transients has been acquired, these will further be cut down to determine what alerts of the sample are supernova candidates, this will be done using the SuperNNova and Early SN Ia classifiers within Fink we will cut off samples that have a classification probability that is less than 0.5 along with spot checking of the individual light curves (Möller et al., 2025). If due to time constraints the machine learning classifiers approach is not fruitful, Bazin fits will be undertaken to classify light curves based on their rise and fall times (Bazin et al., 2011).

Once a sample of extragalactic SN candidates have been acquired, we will then use the empirical fitting algorithm SALT3 to derive the SN colour and stretch properties of the SNe. SALT3 will integrate the light curves to not only determine stretch and colour of the SNe Ia but to also model light curves that will be used to find the EEx SNe Ia (Guy et al., 2005). With the SALT3 fitted light curves we will then apply the colour ($-0.3 \leq c \leq 0.3$), stretch ($-3.0 \leq x_1 \leq 3.0$) and g band magnitude cuts ($-19.95 \leq peak_g \leq -18.35$) made in Deckers et al. (2022) to produce a sample of 'normal' SNe Ia. Then the EEx SNe Ia can be found by subtracting the fitted light curve flux from the observed light curve flux and searching for positive residuals in the light curve. This method is adapted from Deckers et al. (2022) to quantitatively identify EEx SNe Ia.

Finally, using the normalised DLR method we will associate the SNe Ia with their host galaxies. The EEx SNe Ia will be correlated with the SFR, mass and global colour of the host galaxy (Gupta et al., 2016). The SFR, mass, normalised DLR and global colour of the hosts will be acquired using existing methods.

7 References

References

Anderson, J. P., James, P. A., Förster, F., González-Gaitán, S., Habergham, S. M., Hamuy, M., & Lyman, J. D. (2015). On the environments of type Ia supernovae within host galaxies [arXiv:1412.6315 [astro-ph]]. *Monthly Notices of the Royal Astronomical Society*, *448*(1), 732–753. https://doi.org/10.1093/mnras/stu2712

Bazin, G., Ruhlmann-Kleider, V., Palanque-Delabrouille, N., Rich, J., Aubourg, E., Astier, P., Balland, C., Basa, S., Carlberg, R. G., Conley, A., Fouchez, D., Guy, J., Hardin, D., Hook, I. M., Howell, D. A., Pain, R., Perrett, K., Pritchet, C. J., Regnault, N., … Walker, E. S. (2011). Photometric selection of Type Ia supernovae in the Supernova Legacy Survey. *Astronomy & Astrophysics*, *534*, A43. https://doi.org/10.1051/0004-6361/201116898

Bianco, F. B., Ivezić, Ž., Jones, R. L., Graham, M. L., Marshall, P., Saha, A., Strauss, M. A., Yoachim, P., Ribeiro, T., Anguita, T., Bauer, A. E., Bauer, F. E., Bellm, E. C., Blum, R. D., Brandt, W. N., Brough, S., Catelan, M., Clarkson, W. I., Connolly, A. J., … Willman, B. (2022). Optimization of the Observing Cadence for the Rubin Observatory Legacy Survey of Space and Time: A Pioneering Process of Community-focused Experimental Design. *The Astrophysical Journal Supplement Series*, *258*(1), 1. https://doi.org/10.3847/1538-4365/ac3e72

Brout, D., Scolnic, D., Popovic, B., Riess, A. G., Carr, A., Zuntz, J., Kessler, R., Davis, T. M., Hinton, S., Jones, D., Kenworthy, W. D., Peterson, E. R., Said, K., Taylor, G., Ali, N., Armstrong, P., Charvu, P., Dwomoh, A., Meldorf, C., … Wiseman, P. (2022). The Pantheon+ Analysis: Cosmological Constraints. *The Astrophysical Journal*, *938*(2), 110. https://doi.org/10.3847/1538-4357/ac8e04

Burke, J., Howell, D. A., Sand, D. J., Amaro, R. C., Brown, P. J., Andrews, J. E., Bostroem, K. A., Dong, Y., Haislip, J., Hiramatsu, D., Hosseinzadeh, G., Kouprianov, V., Lundquist, M. J., McCully, C., Pellegrino, C., Reichart, D., Tartaglia, L., Valenti, S., & Yang, S. (2022, July). Early Lightcurves of Type Ia Supernovae are Consis-

tent with Nondegenerate Progenitor Companions [arXiv:2207.07681 [astro-ph]]. https://doi.org/10.48550/arXiv.2207.07681

Deckers, M., Maguire, K., Magee, M. R., Dimitriadis, G., Smith, M., Murieta, A. S. d., Miller, A. A., Goobar, A., Nordin, J., Rigault, M., Bellm, E., Coughlin, M. W., Laher, R. R., Shupe, D., Graham, M. J., Kasliwal, M. M., & Walters, R. (2022). Constraining Type Ia supernova explosions and early flux excesses with the Zwicky Transient Factory [arXiv:2202.12914 [astro-ph]]. *Monthly Notices of the Royal Astronomical Society*, *512*(1), 1317–1340. https://doi.org/10.1093/mnras/stac558

Ginolin, M., Rigault, M., Smith, M., Copin, Y., Ruppin, F., Dimitriadis, G., Goobar, A., Johansson, J., Maguire, K., Nordin, J., Amenouche, M., Aubert, M., Barjou-Delayre, C., Betoule, M., Burgaz, U., Carreres, B., Deckers, M., Dhawan, S., Feinstein, F., … Sharma, Y. (2025). ZTF SN Ia DR2: Environmental dependencies of stretch and luminosity of a volume limited sample of 1,000 Type Ia Supernovae [arXiv:2405.20965 [astro-ph]]. *Astronomy & Astrophysics*, *695*, A140. https://doi.org/10.1051/0004-6361/202450378

Gupta, R. R., Kuhlmann, S., Kovacs, E., Spinka, H., Kessler, R., Goldstein, D. A., Liotine, C., Pomian, K., D'Andrea, C. B., Sullivan, M., Carretero, J., Castander, F. J., Nichol, R. C., Finley, D. A., Fischer, J. A., Foley, R. J., Kim, A. G., Papadopoulos, A., Sako, M., … Wester, W. (2016). Host Galaxy Identification for Supernova Surveys [arXiv:1604.06138 [astro-ph]]. *The Astronomical Journal*, *152*(6), 154. https://doi.org/10.3847/0004-6256/152/6/154

Guy, J., Astier, P., Nobili, S., Regnault, N., & Pain, R. (2005). SALT: A spectral adaptive light curve template for type Ia supernovae. *Astronomy & Astrophysics*, *443*(3), 781–791. https://doi.org/10.1051/0004-6361:20053025

Hoyle, F., & Fowler, W. A. (1960). Nucleosynthesis in Supernovae. [ADS Bibcode: 1960ApJ...132..565H]. *The Astrophysical Journal*, *132*, 565. https://doi.org/10.1086/146963

Iben, I., Jr., & Tutukov, A. V. (1984). Supernovae of type I as end products of the evolution of binaries with components of moderate initial mass. [ADS Bibcode: 1984ApJS...54..335I]. *The Astrophysical Journal Supplement Series*, *54*, 335–372. https://doi.org/10.1086/190932

Kasen, D. (2010). Seeing the Collision of a Supernova with its Companion Star [arXiv:0909.0275 [astro-ph.HE]]. *The Astrophysical Journal*, *708*(2), 1025–1031. https://doi.org/10.1088/0004-637X/708/2/1025

Kelly, P. L., Hicken, M., Burke, D. L., Mandel, K. S., & Kirshner, R. P. (2010). Hubble Residuals of Nearby Type Ia Supernovae Are Correlated with Host Galaxy Masses [arXiv:0912.0929 [astro-ph]]. *The Astrophysical Journal*, *715*(2), 743–756. https://doi.org/10.1088/0004-637X/715/2/743

Kelsey, L., Sullivan, M., Wiseman, P., Armstrong, P., Chen, R., Brout, D., Davis, T. M., Dixon, M., Frohmaier, C., Galbany, L., Graur, O., Kessler, R., Lidman, C., Möller, A., Popovic, B., Rose, B., Scolnic, D., Smith, M., Vincenzi, M., … Weaverdyck, N. (2022). Concerning Colour: The Effect of Environment on Type Ia Supernova Colour in the Dark Energy Survey [arXiv:2208.01357 [astro-ph]]. *Monthly Notices of the Royal Astronomical Society*, *519*(2), 3046–3063. https://doi.org/10.1093/mnras/stac3711

Kenworthy, W. D., Jones, D. O., Dai, M., Kessler, R., Scolnic, D., Brout, D., Siebert, M. R., Pierel, J. D. R., Dettman, K. G., Dimitriadis, G., Foley, R. J., Jha, S. W., Pan, Y.-C., Riess, A., Rodney, S., & Rojas-Bravo, C. (2021). SALT3: An Improved Type Ia Supernova Model for Measuring Cosmic Distances. *The Astrophysical Journal*, *923*(2), 265. https://doi.org/10.3847/1538-4357/ac30d8

Lampeitl, H., Smith, M., Nichol, R. C., Bassett, B., Cinabro, D., Dilday, B., Foley, R. J., Frieman, J. A., Garnavich, P. M., Goobar, A., Im, M., Jha, S. W., Marriner, J., Miquel, R., Nordin, J., Östman, L., Riess, A. G., Sako, M., Schneider, D. P., … Stritzinger, M. (2010). THE EFFECT OF HOST GALAXIES ON TYPE Ia SUPERNOVAE IN THE SDSS-II SUPERNOVA SURVEY. *The Astrophysical Journal*, *722*(1), 566–576. https://doi.org/10.1088/0004-637X/722/1/566

Leoni, M., Ishida, E. E. O., Peloton, J., & Möller, A. (2022). Fink: Early supernovae Ia classification using active learning [arXiv:2111.11438 [astro-ph]]. *Astronomy & Astrophysics*, *663*, A13. https://doi.org/10.1051/0004-6361/202142715

Lim, G., Im, M., Paek, G. S. H., Yoon, S.-C., Choi, C., Kim, S., Wheeler, J. C., Thomas, B. P., Vinkó, J., Kim, D., Seo, J., Kang, W., Kim, T., Sung, H.-I., Kim, Y., Yoon, J.-N., Kim, H., Kim, J., Bae, H., … Mirzaqulov, D. (2023). The Early Light Curve of a Type

Ia Supernova 2021hpr in NGC 3147: Progenitor Constraints with the Companion Interaction Model [arXiv:2303.05051 [astro-ph]]. *The Astrophysical Journal*, *949*(1), 33. https://doi.org/10.3847/1538-4357/acc10c

The LSST Camera (LSSTCam). (n.d.). Retrieved May 11, 2026, from https://lsstcam.lsst.io/index.html

Magee, M. R., Sim, S. A., Kotak, R., & Kerzendorf, W. E. (2018). Modelling the early time behaviour of type Ia supernovae: Effects of the^{56} Ni distribution. *Astronomy & Astrophysics*, *614*, A115. https://doi.org/10.1051/0004-6361/201832675

Marion, G. H., Brown, P. J., Vinkó, J., Silverman, J. M., Sand, D. J., Challis, P., Kirshner, R. P., Wheeler, J. C., Berlind, P., Brown, W. R., Calkins, M. L., Camacho, Y., Dhungana, G., Foley, R. J., Friedman, A. S., Graham, M. L., Howell, D. A., Hsiao, E. Y., Irwin, J. M., … Zheng, W. (2016). SN~2012cg: Evidence for Interaction Between a Normal Type Ia Supernova and a Non-Degenerate Binary Companion [arXiv:1507.07261 [astro-ph]]. *The Astrophysical Journal*, *820*(2), 92. https://doi.org/10.3847/0004-637X/820/2/92

Martin, B., Lidman, C., Brout, D., Tucker, B. E., Dixon, M., & Armstrong, P. (2024). [O ii] as an effective indicator of the dependence between the standardized luminosities of Type Ia supernovae and the properties of their host galaxies. *Monthly Notices of the Royal Astronomical Society*, *533*(3), 2640–2655. https://doi.org/10.1093/mnras/stae1996

Möller, A., & de Boissière, T. (2020). SuperNNova: An open-source framework for Bayesian, neural network-based supernova classification. *Monthly Notices of the Royal Astronomical Society*, *491*(3), 4277–4293. https://doi.org/10.1093/mnras/stz3312

Möller, A., Ishida, E. E. O., Peloton, J., Velázquez, O. V., Soon, J., Martin, B., Cluver, M., Leoni, M., & Taylor, E. (2025). Real-Time Active Learning for optimised spectroscopic follow-up: Enhancing early SN Ia classification with the Fink broker [arXiv:2502.19555 [astro-ph]]. *Publications of the Astronomical Society of Australia*, *42*, e057. https://doi.org/10.1017/pasa.2025.20

Möller, A., Peloton, J., Ishida, E. E. O., Arnault, C., Bachelet, E., Blaineau, T., Boutigny, D., Chauhan, A., Gangler, E., Hernandez, F., Hrivnac, J., Leoni, M., Leroy, N., Moniez,

M., Pateyron, S., Ramparison, A., Turpin, D., Ansari, R., Allam, T., … Webb, N. (2021). Fink, a new generation of broker for the LSST community [arXiv:2009.10185 [astro-ph]]. *Monthly Notices of the Royal Astronomical Society*, *501*(3), 3272–3288. https://doi.org/10.1093/mnras/staa3602

Perlmutter, S., Aldering, G., Goldhaber, G., Knop, R. A., Nugent, P., Castro, P. G., Deustua, S., Fabbro, S., Goobar, A., Groom, D. E., Hook, I. M., Kim, A. G., Kim, M. Y., Lee, J. C., Nunes, N. J., Pain, R., Pennypacker, C. R., Quimby, R., Lidman, C., … Project, T. S. C. (1999). Measurements of and from 42 High-Redshift Supernovae. *The Astrophysical Journal*, *517*(2), 565. https://doi.org/10.1086/307221

Ramaiya, S., Vincenzi, M., Jarvis, M. J., Wiseman, P., & Sullivan, M. (2025, September). The dependence of the Type Ia Supernova colour-luminosity relation on their host galaxy properties [arXiv:2509.12035 [astro-ph]]. https://doi.org/10.48550/arXiv.2509.12035

Riess, A. G., Filippenko, A. V., Challis, P., Clocchiatti, A., Diercks, A., Garnavich, P. M., Gilliland, R. L., Hogan, C. J., Jha, S., Kirshner, R. P., Leibundgut, B., Phillips, M. M., Reiss, D., Schmidt, B. P., Schommer, R. A., Smith, R. C., Spyromilio, J., Stubbs, C., Suntzeff, N. B., & Tonry, J. (1998). Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *The Astronomical Journal*, *116*(3), 1009–1038. https://doi.org/10.1086/300499

Rigault, M., Brinnel, V., Aldering, G., Antilogus, P., Aragon, C., Bailey, S., Baltay, C., Barbary, K., Bongard, S., Boone, K., Buton, C., Childress, M., Chotard, N., Copin, Y., Dixon, S., Fagrelius, P., Feindt, U., Fouchez, D., Gangler, E., … Thomas, R. C. (2020). Strong Dependence of Type Ia Supernova Standardization on the Local Specific Star Formation Rate [arXiv:1806.03849 [astro-ph]]. *Astronomy & Astrophysics*, *644*, A176. https://doi.org/10.1051/0004-6361/201730404

Sullivan, M., Conley, A., Howell, D. A., Neill, J. D., Astier, P., Balland, C., Basa, S., Carlberg, R. G., Fouchez, D., Guy, J., Hardin, D., Hook, I. M., Pain, R., Palanque-Delabrouille, N., Perrett, K. M., Pritchet, C. J., Regnault, N., Rich, J., Ruhlmann-Kleider, V., … Walker, E. S. (2010). The dependence of Type Ia Supernovae luminosities on their host galaxies: SN Ia host galaxies. *Monthly Notices of the Royal Astronomical Society*, no–no. https://doi.org/10.1111/j.1365-2966.2010.16731.x

Toy, M., Wiseman, P., Sullivan, M., Scolnic, D., Vincenzi, M., Brout, D., Davis, T. M., Frohmaier, C., Galbany, L., Lidman, C., Lee, J., Kelsey, L., Kessler, R., Möller, A., Popovic, B., Sánchez, B. O., Shah, P., Smith, M., Allam, S., … Weaverdyck, N. (2025, March). Reduction of the type Ia supernova host galaxy step in the outer regions of galaxies [arXiv:2408.03749 [astro-ph]]. https://doi.org/10.48550/arXiv.2408.03749

Tripp, R. (1998). A two-parameter luminosity correction for Type Ia supernovae?

Vera C. Rubin Observatory Survey Strategy. (n.d.). Retrieved May 8, 2026, from https://survey-strategy.lsst.io/index.html

Whelan, J., & Iben, I., Jr. (1973). Binaries and Supernovae of Type I. *The Astrophysical Journal*, *186*, 1007. https://doi.org/10.1086/152565

Wiseman, P., Sullivan, M., Smith, M., & Popovic, B. (2023). Further evidence that galaxy age drives observed type Ia supernova luminosity differences [arXiv:2302.05341 [astro-ph]]. *Monthly Notices of the Royal Astronomical Society*, *520*(4), 6214–6222. https://doi.org/10.1093/mnras/stad488

Wu, W., Jiang, J.-a., Xu, Z., Meng, D., Maeda, K., Kuncarayakti, H., Galbany, L., Jha, S. W., Ivezić, Ž., Yoachim, P., Liu, Z., Zhao, J., Wang, T., Kong, X., Connolly, A. J., Jia, Z., Hu, L., Fan, L., Jiang, N., … Zhu, Q. (2026, February). WFST Supernovae in the First Year: I. Statistical Study of 16 Early-phase Type Ia Supernovae from the Pilot Survey [arXiv:2602.17268 [astro-ph]]. https://doi.org/10.48550/arXiv.2602.17268